

Valorization of marine resources

Strategic Analysis of Fisheries and Aquaculture in Thailand: Challenges and Sustainable Prospects

Country profile: Thailand



Map of Thailand showing its EEZs (Janekitkosol et al., 2003).

Location :

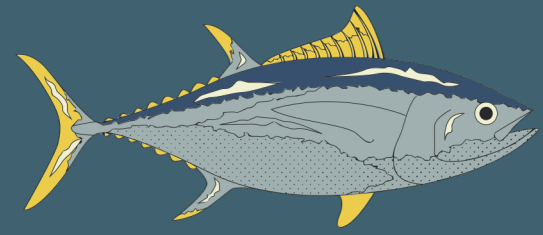
Southeast Asia
Indochinese Peninsula

Economy:

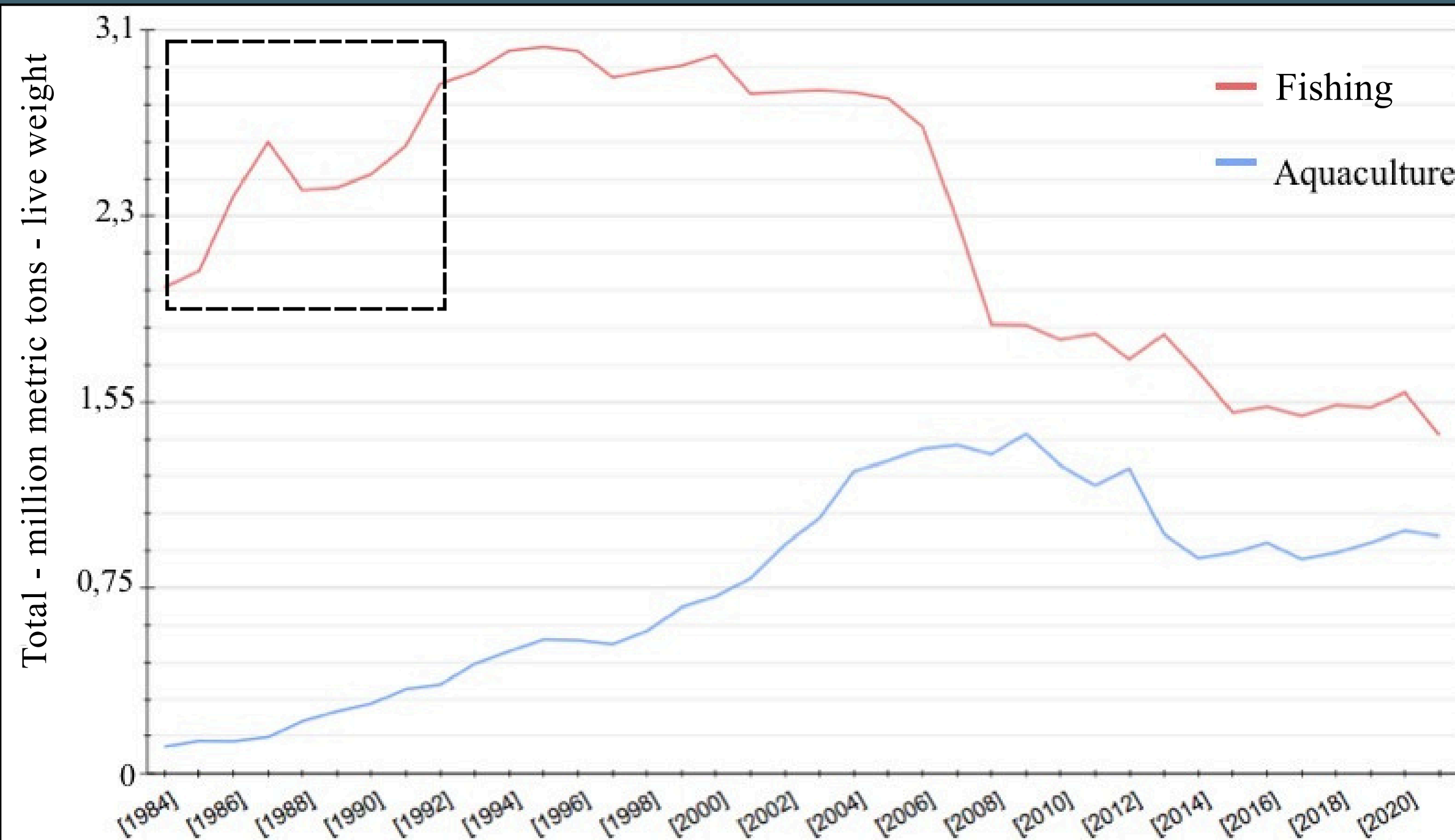
- GDP: \$495 billion
- 2nd largest economy in Southeast Asia
- 71 million people
- GDP per capita: \$6,909
- Dependent on tourism and exportations

Marine areas:

2 EEZ: 420 280 km²
Coast: 3 219 km



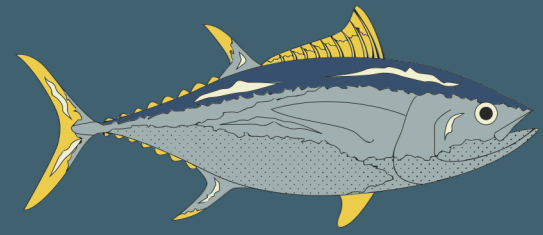
Thai productions and their evolution



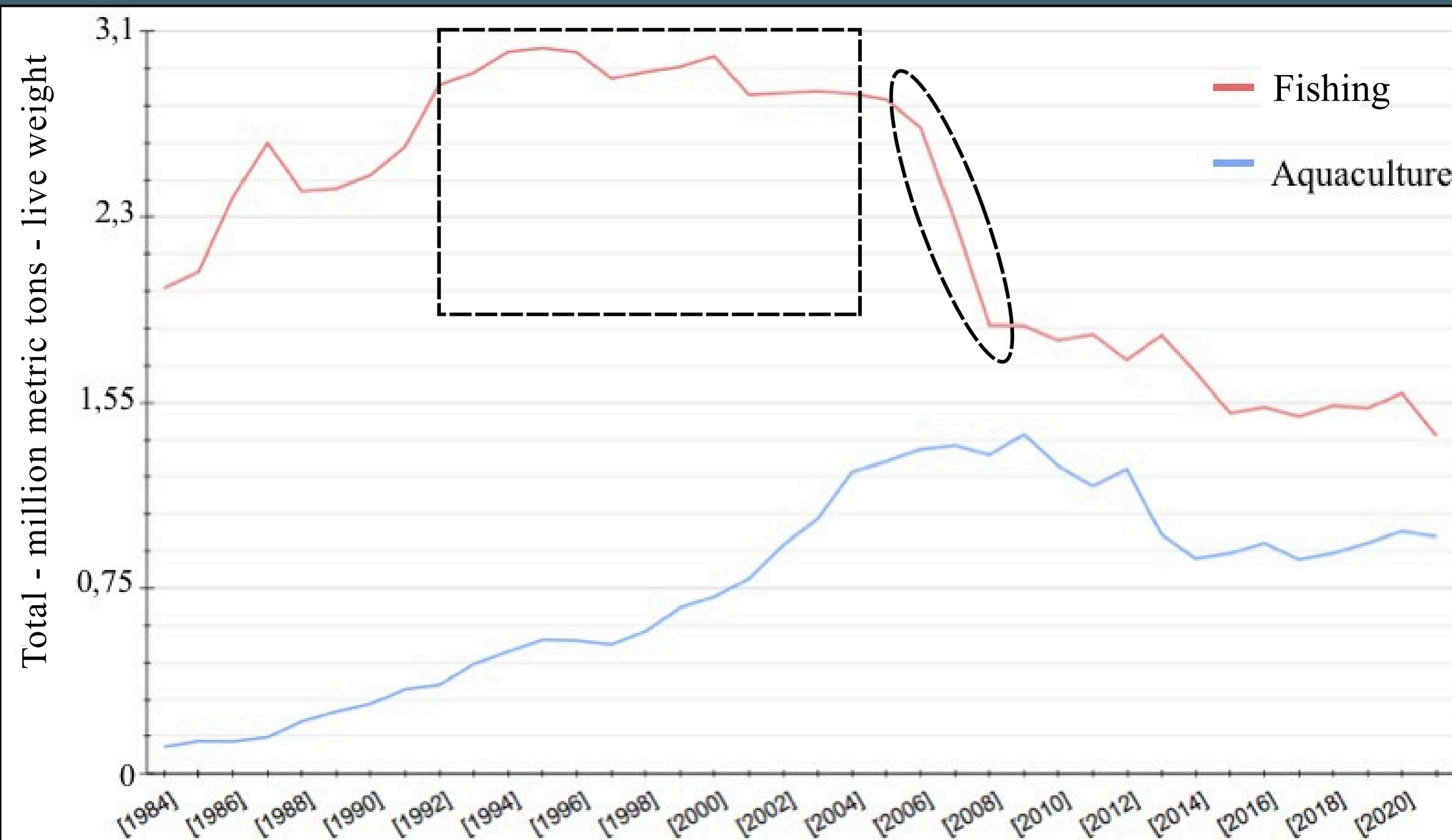
Graph showing the trend in total catch from fisheries (red) and aquaculture (blue) in Thailand from 1984 to 2021, in metric tons of live weight. (Data: FAO, 2023)

Fishing :

- Until 1990: development



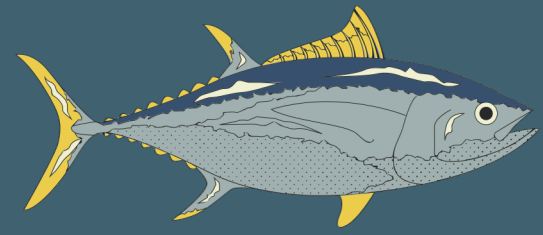
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Fishing :

- Until 1990: development
- From 1992 to 2004: stagnant catch levels = **depleted stocks**
- 2005: **collapse** in stock levels



Thai productions and their evolution

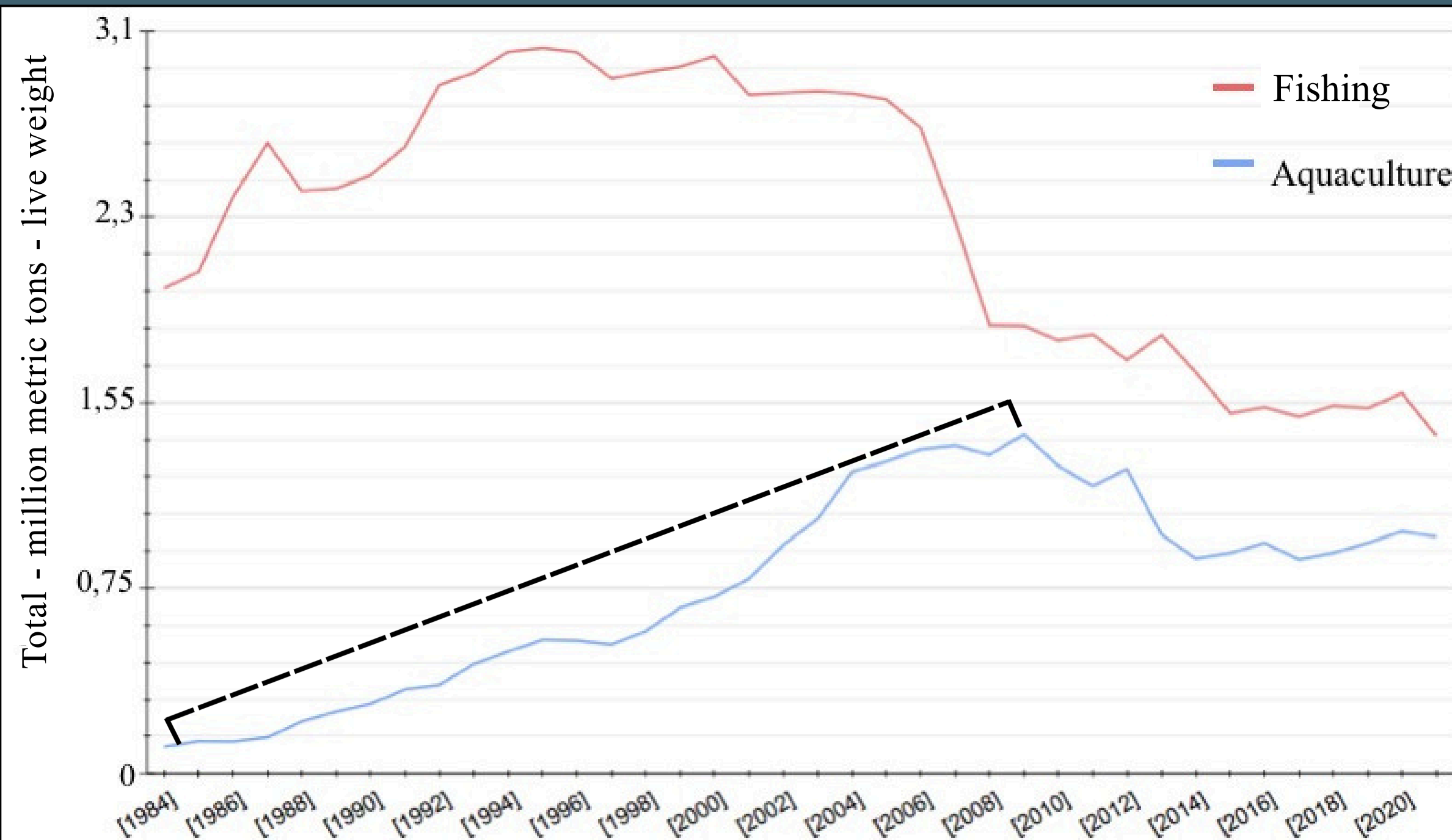


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Fishing :

- Until 1990: development
- From 1992 to 2004: stagnant catch levels = **depleted stocks**
- 2005: **collapse** in stock levels
- Before 2015: unrestricted fishing = **overfishing**
- 2020: **12th**-largest producer

Thai productions and their evolution

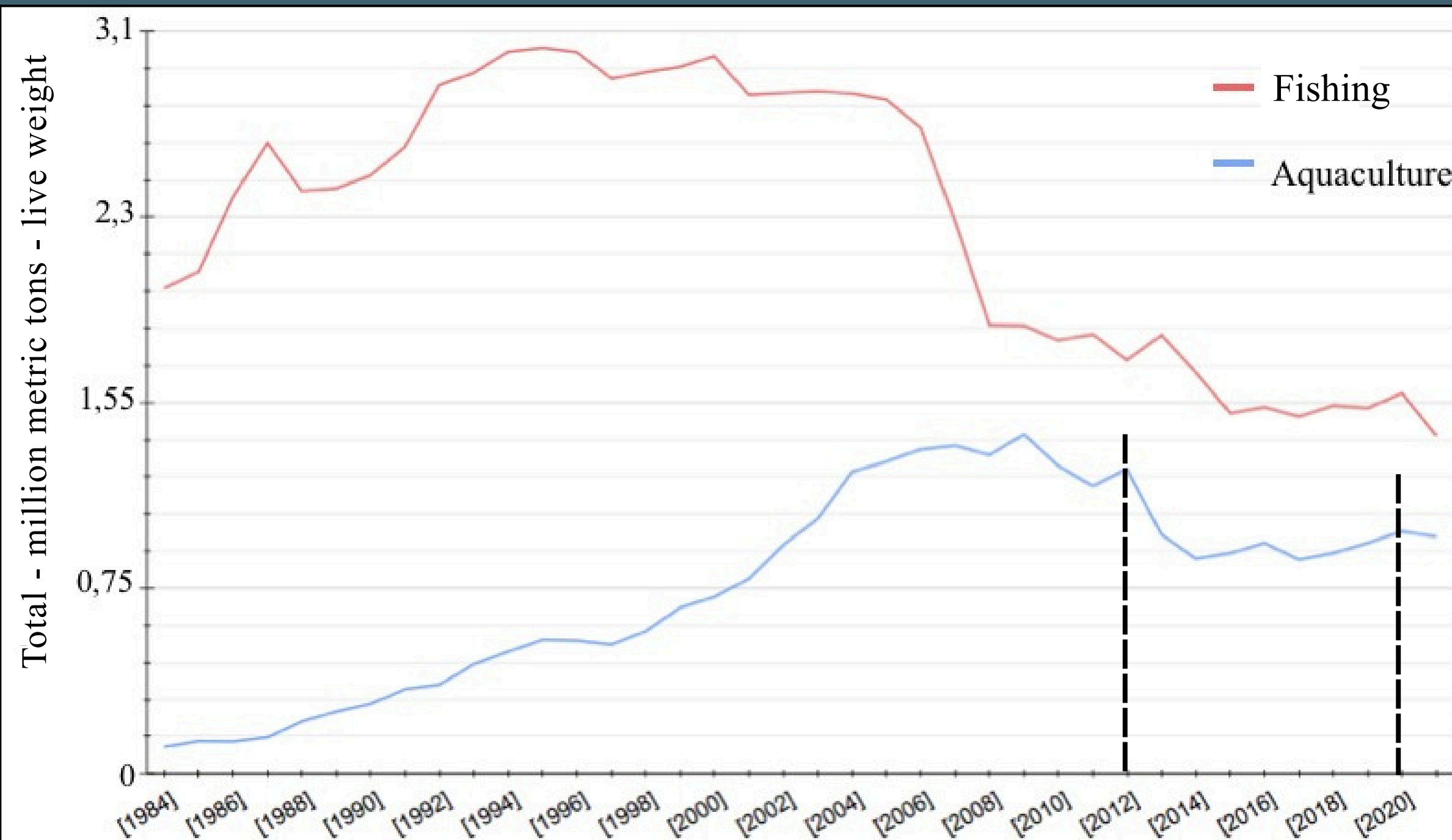


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Aquaculture:

- 1970: beginning of aquaculture
- Development: more species

Productions thaïlandaises et leurs évolutions



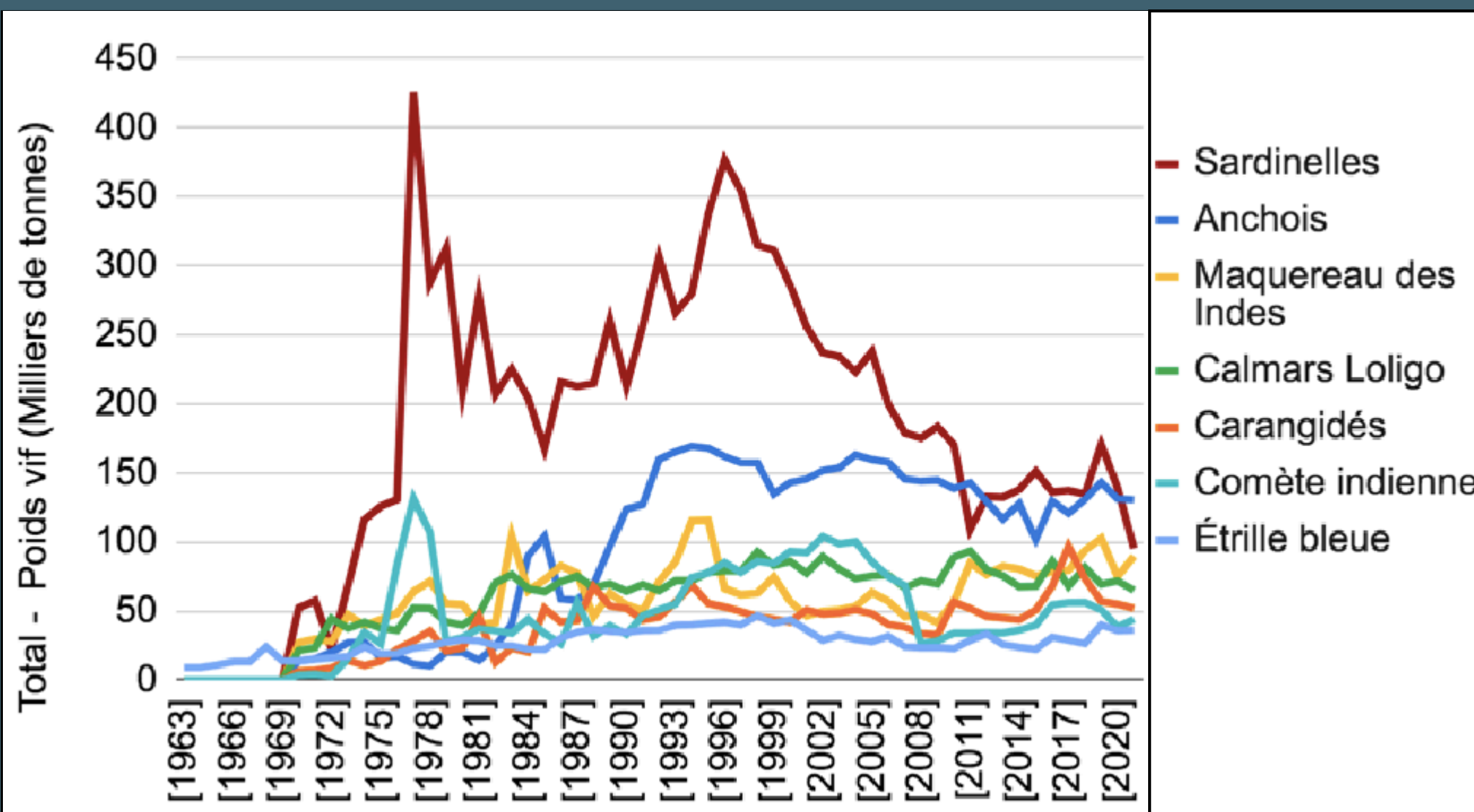
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Aquaculture:

- 1970: beginning of aquaculture
- Development: more species
- 2012: early mortality syndrome (EMS)
- 2020: **6th**-largest producer of aquacultured crustaceans



Breakdown of production: fishing



Total catches (in thousands of metric tons of live weight) of the seven most commonly caught species by the Thai fishing fleet from 1963 to 2021. (Data: FAO)

- Pelagic fish: **74,8%**

- anchovies (25%)
- sardines (18%)
- mackerel (17%)

- Since 1980:

- ↘ sardines (-32%) → ↘ **stocks**
- ↗ anchovies (+ 20%) → **techniques**
- **Overfished** pelagic resources

- 2020:

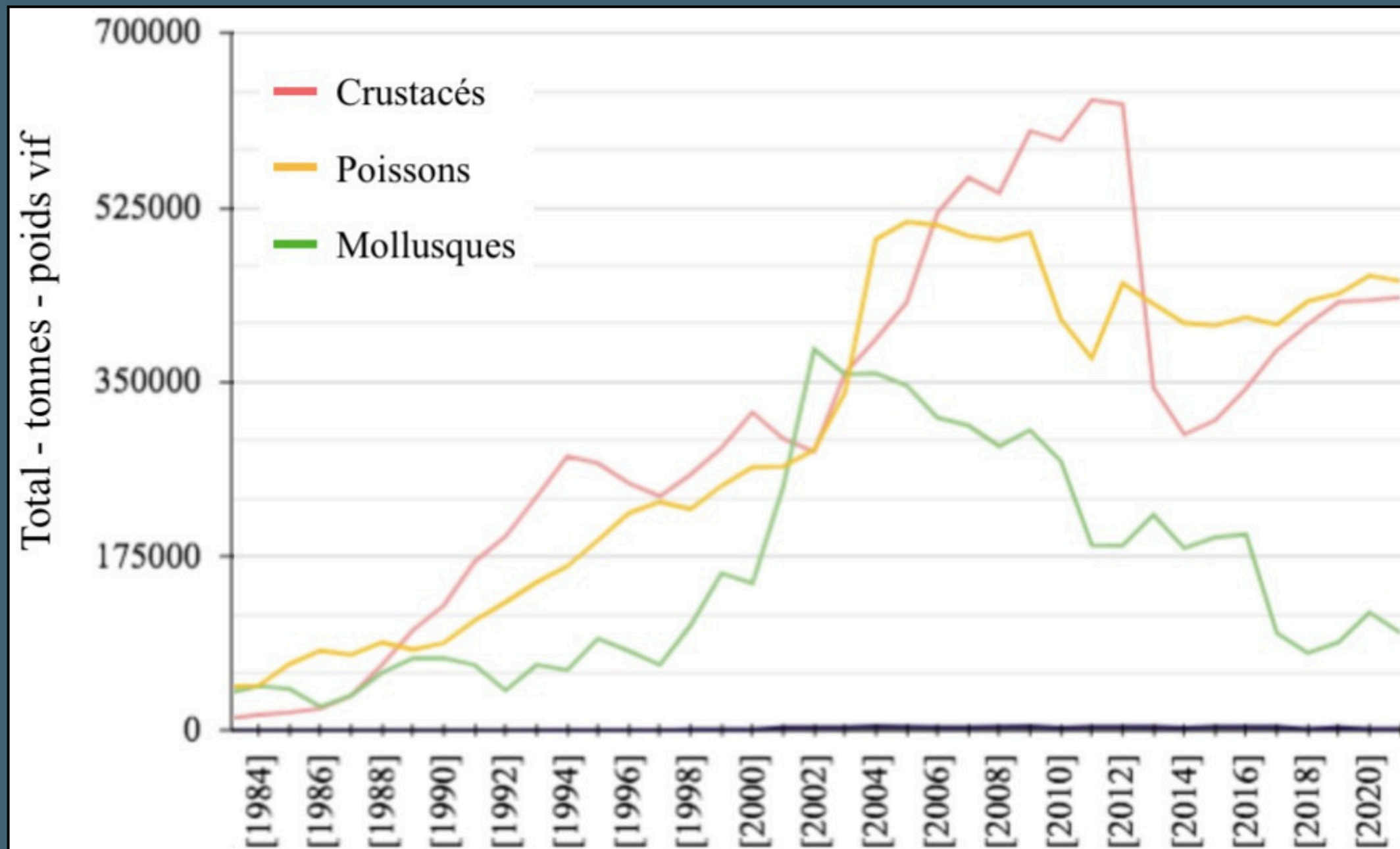
- The **6th**-largest fishing nation for herring, sardines, and anchovies

Breakdown of production: aquaculture



Coastal
waters: 62%

Freshwater:
38%



Trends in the production of Thailand's major aquaculture products from 1970 to 2020.
(Data: FAO, 2023)

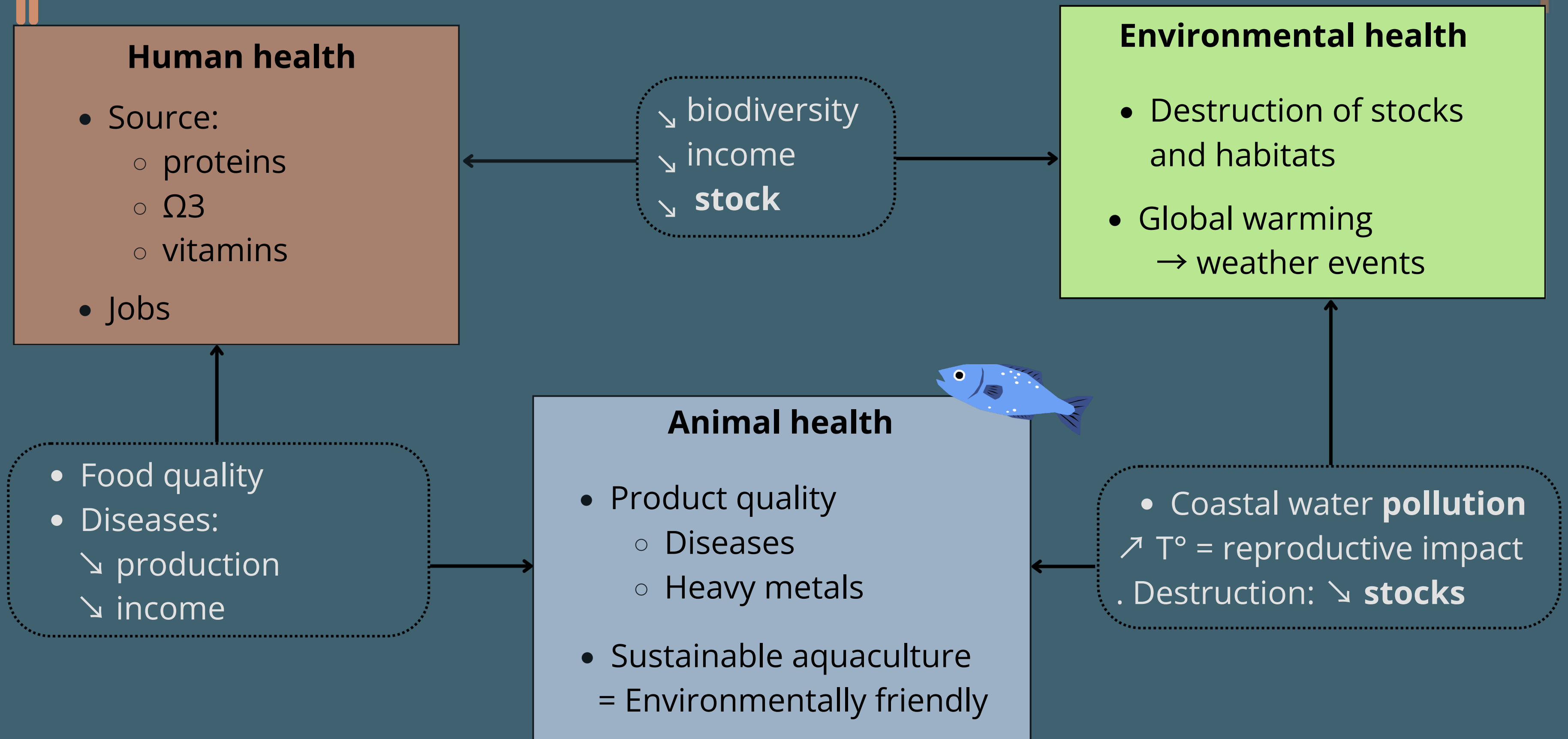
Key products in 2022:

- Crustaceans (40%)
 - 90% shrimps
 - 95% in coastal waters
- Fishes (38%)
 - 96% in freshwater
- Molluscs (22%)

En 2021:

n°1 fishes → 452,000 metric tons
n°2 shrimps → 435,000 metric tons

Links between fishing and food security



Market Organization

The supply chain in 6 segments:

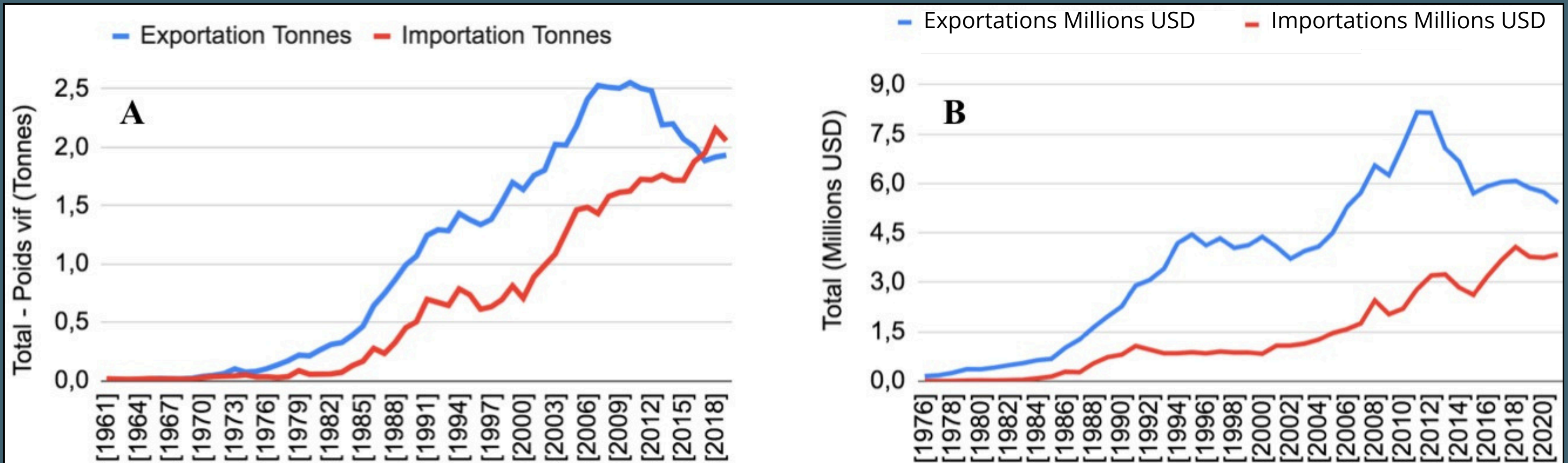
1. Primary suppliers (fishing and aquaculture)
2. Middlemen (fish market)
3. Processors
4. Distributors (exporters and importers)
5. Retailers (supermarkets)
6. Consumers

Suite aux différentes crises depuis 1990:

- ↘ profits
- ↗ costs (processing, quality control, etc.)
- Registered employees
- Labor costs:
 - 20% of the total cost
 - > to neighboring countries:
 - \$9.85/day
 - \$5.75/day in Vietnam

**→ Loss of competitiveness
compared to other Asian producers**

Market Organization



Charts showing total exports and imports of aquatic products in tons from 1961 to 2018 (A) and in millions of U.S. dollars from 1976 to 2020 (B). (Source: FAO, 2023)

- Highly **offensive** exporter in 2012
- 2012: ↘ in exports = **shrimp virus**
- Imports > exports → growth of the **tuna canning** industry
- Exports generate more revenue than imports → \$5.86 million vs. \$3.77 million

Market Organization

Exports:

62% Pelagic fish

16% Shrimp

7% Mollusks

6% Demersal fish

5.4% Freshwater fish

2.8% Cephalopods

Imports:

70% Pelagic fish

10% Cephalopods

8.6% Freshwater fish

4.7% Mollusks

3.9% Crustaceans

Thuna :

Leader in the canned tuna industry

Thai Union holds a 37% market share

Imports 80% of the tuna it uses



Shrimp:

Asia's leading exporter

→ The United States, Europe, and Asia (China)
Strong competition from neighboring countries
(China, India)

\$1.87 billion in 2021



Fishing model

In 2020:

- 172,430 fishermen; 80% of aquaculture
- production comes from family-run farms

Thai fleet:

- **Multi-gear** and **multi-species** fishing
- In 2015: 42,512 vessels were recorded
 - 78% small **craft**
 - 22% **commercial** vessels

Fishing methods:

- Commercial vessels: **90%** of the total catch
 - 41% **trawlers** (medium or large)
 - 14% encircling gillnets
 -
- Traditional boats:
 - Small trawls, gillnets
 - Traps, lines, and longlines.

Fishing system:

- Overfishing → decline in fish stocks
- Implementation of **licenses** and **quotas**
 - fishing and aquaculture model
→ **more sustainable and environmentally friendly**



National policies

Since 1984:

- Regulations: fishing zones, gear used, and number of vessels
- **Seasonal** closure of certain **fishing zones**
- Prohibition of trawls and seines with a mesh size of less than 4.7 cm in certain zones
--> Protection of juveniles and other species (mackerel)

Problem: no enforcement by authorities + no permit management = overfishing

Since 2015: Royal Decree on Fishing

- Limits on the **number of licenses** issued
- Compliance with designated **zones** and **seasons** + use of **appropriate equipment**
- **Combating** illegal fishing
- Ban on the construction of new aquaculture farms + operating permits
- Mandatory **monitoring** of aquaculture operations (FAO Code of Conduct)

Goal: to align with the United Nations Sustainable Development Goals



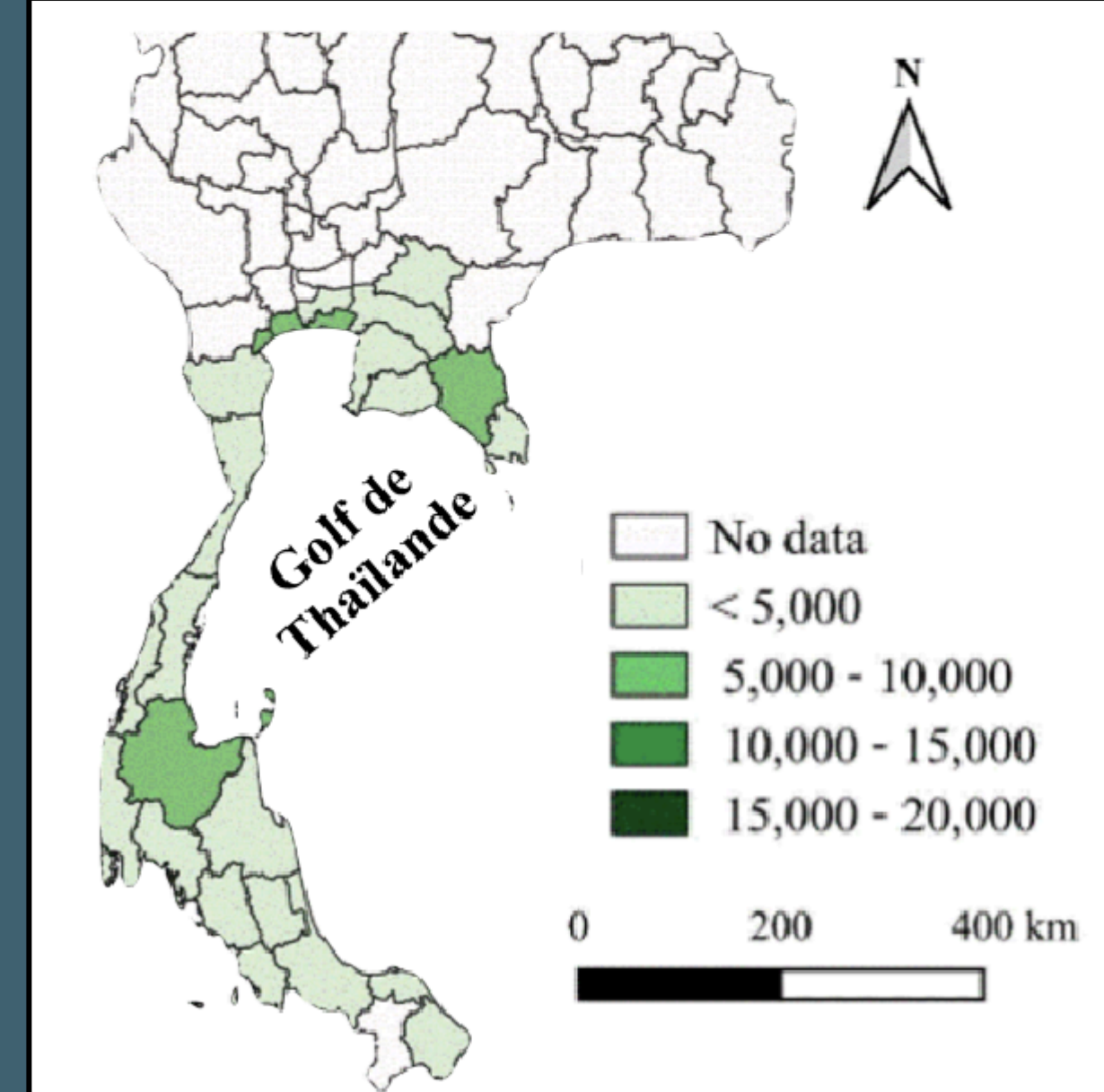
Regional impact

Fishing :

- Destruction:
 - seabed, fish stocks, biodiversity
- Decline in catches

Aquaculture:

- Destruction of mangroves (spawning grounds)
 - home to a rich **biodiversity** and serve as a natural
 - barrier against **erosion** and **flooding**
- Destruction of habitats /
- biodiversity **water pollution**



Map of Thailand showing the area in hectares of land used for coastal aquaculture in 2015 (Sampantamit et al., 2020).

SWOT Assessment:

STRENGTHS

- Geographical location, climate, biodiversity
- Multi-gear fishing
- Wide **variety** of organisms
- Rivers = aquaculture
- Infrastructure developed in **coastal areas**
- Government policies support **sustainable** and environmentally responsible practices
- ↗ local demand for **protein**

WEAKNESSES

- **Destruction** of coastal habitats
- Susceptible to **disease**, focused on shrimp
- **Declining stocks**
- **Limited** market access in rural areas
- Lack of enforcement and oversight of **regulations**
- Tough working conditions, accidents
- commercial ships = **competition**

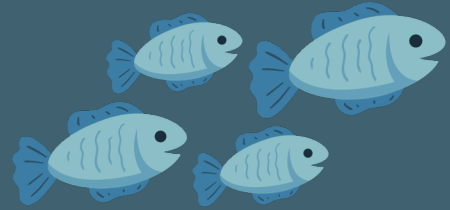
OPPORTUNITIES

SWOT Assessment:

THREATS

- Developing (sustainable) aquaculture
 - **Protecting** the environment
 - Recirculating aquaculture systems
- Seafood = a **source of protein**
 - ↗ global **demand**
 - Aquaculture: developing high-value-added products (**seaweed**)
- Raise awareness among fishermen and aquaculturists about **best practices**
- Strengthen existing **regulations**
 - Implement **traceability** and quality assurance (MSC) —> boost competitiveness

- Global warming
- Strong **competition** from neighboring countries with cheaper labor
- Export-dependent economy
- **Competitor** countries' dependence on raw material supplies
- International regulations = **challenges** for aquaculturists and fishermen



Conclusion



Fishing model based on overfishing →
multi-gear and **multi-species**

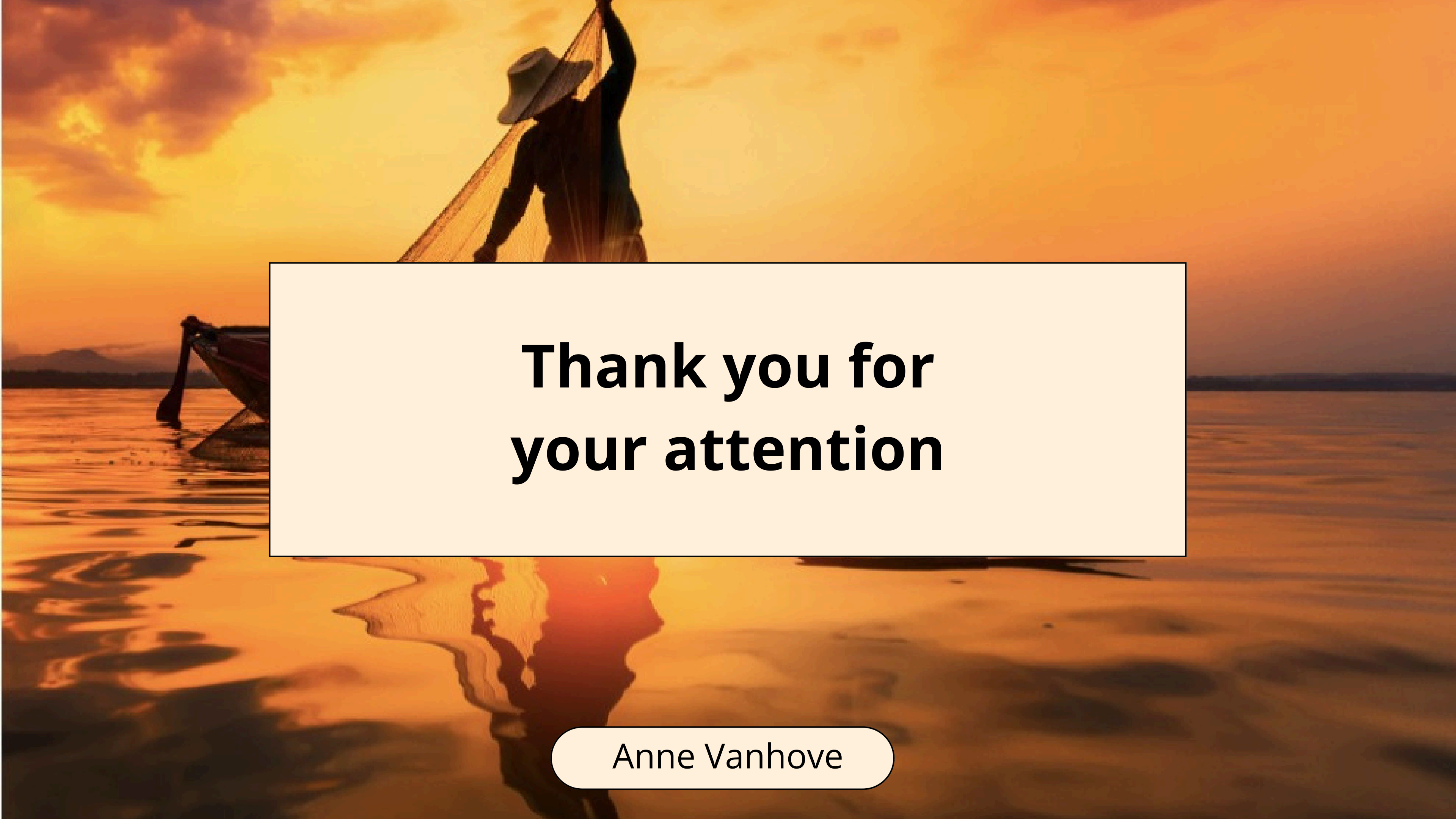
- Developing more sustainable aquaculture and fishing practices
 - requires financial assistance to train
 - aquaculturists and fishermen

- Aquaculture = protein
 - ↗ Global demand and the potential
 - for diversification = ↗ Profits from
 - integrated multi-trophic aquaculture
 - disease control + ↗ nutritional
 - value, cost-effective, and affordable

- Ensure **compliance with applicable regulations**
 - restoring fish stocks
 - ↘ forced labor
 - = ↗ jobs for Thais

- Implementation of quality control and traceability
 - producing high-quality products that are in demand on the global market

Fishing and aquaculture play a central role in the country's economy and food security.

A fisherman wearing a wide-brimmed hat is silhouetted against a vibrant sunset sky, pulling a large fishing net. The scene is reflected in the calm water below. A small boat is visible on the left side of the frame.

**Thank you for
your attention**

Anne Vanhove

Sources

- F. A. O (2023). Aquaculture software. FishStatJ—Software for Fishery and Aquaculture Statistical Time Series. FAO Fisheries Division. Licence CC BY-NC-SA 3.0 IGO. Date de consultation: 02.04.2024.
- Janekitkosol, W., Somchanakij, H., Eiamsa-ard, M., & Supongpan, M. (2003). Strategic review of the fishery situation in Thailand
- Sampantamit, T., Ho, L., Lachat, C., Sutummawong, N., Sorgeloos, P., & Goethals, P. (2020). Aquaculture production and its environmental sustainability in Thailand: Challenges and potential solutions. Sustainability, 12(5), 2010.